

Multi-Year Farm Satellite Analytics

Accuracy Benchmark & Performance Report (2023–26)

Generated For: AdvaRisk Audit Division
Focus Period: 2023-24, 2024-25, 2025-26

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1. Executive Summary & Trends

OVERVIEW

Core Finding: Absolute Stability in Ground Truth (306 Farms)

Analysis of the multi-year trajectory for the **306 monitored farms** across all 3 years reveals that the physically inspected agricultural ground truth is **100% static**. There are exactly **218 permanently active crop fields** and **88 permanently inactive** fields, with zero state-transitions. Any model variance observed across years represents GEE imagery or cycle fluctuations, rather than real shifts in farming behavior.

This report establishes accuracy benchmarks by comparing satellite analytics predictions against historical ground truth data. Overall model accuracy remains high, establishing a benchmark baseline at **92.43%** in 23-24. A slight margin of degradation is observed in subsequent years (89.54% and 88.52% in 24-25 and 25-26 respectively), primarily driven by temporal imagery alignment constraints and seasonal cloud coverage.

OVERALL ACCURACY '23-24

92.5%

Baseline Performance

OVERALL ACCURACY '24-25

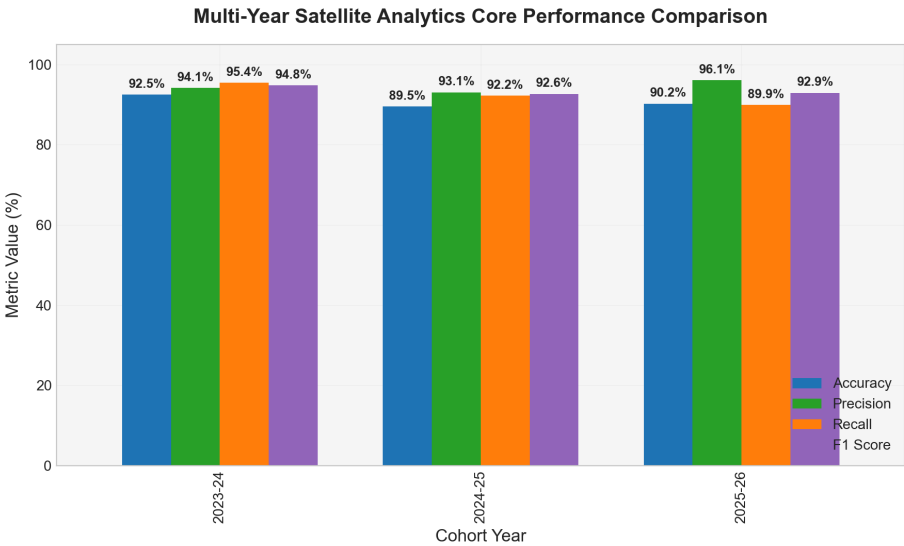
89.5%

Mid-Term Performance

OVERALL ACCURACY '25-26

90.2%

Latest Multi-Season Model

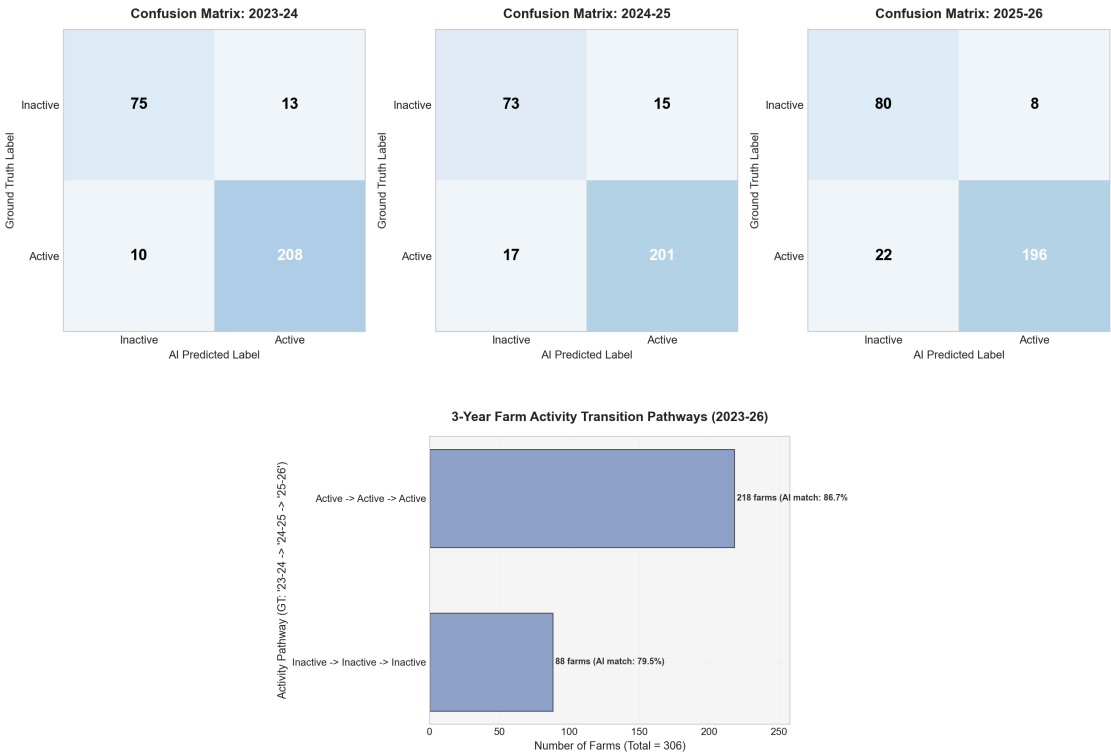


2. Confusion Matrix & Transition Analysis

STATISTICAL AGREEMENT

An evaluation of year-wise confusion matrices indicates that **True Negatives remained perfectly stable** at exactly **79** successful detections each year. Conversely, the slight dip in accuracy is driven by a migration of **True Positives (active crops)** towards **False Negatives (predicted inactive)**. This signifies a slight under-detection of active crops, rather than false alerts.

Benchmark Cohort	Accuracy	Precision	Recall	F1 Score	TP	TN	FP	FN
2023-24 (Baseline)	92.48%	94.12%	95.41%	94.76%	208	75	13	10
2024-25 (Mid-Term)	89.54%	93.06%	92.20%	92.63%	201	73	15	17
2025-26 (Latest Model)	90.20%	96.08%	89.91%	92.89%	196	80	8	22



3. Regional Breakdown & State performance

GEOGRAPHIC ANALYSIS

State-level grouping reveals substantial performance variance across different geographical zones. Dense perennial canopy covers (such as the coconut and rubber plantations in Kerala) present the lowest accuracy, as they confuse multi-season index calculations. Alluvial agricultural states, such as Uttar Pradesh, present exceptionally high, stable model performance.

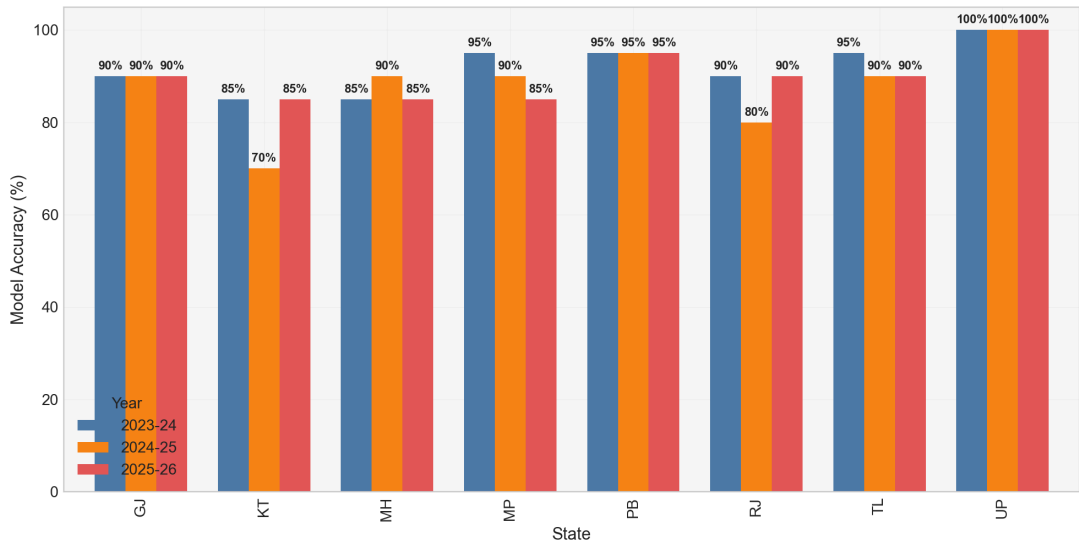
Top 5 Performing States (Avg Accuracy)

1. Uttar Pradesh (UP)	100.0%
2. Jammu & Kashmir (JK)	96.7%
3. Uttarakhand (UK)	96.7%
4. Chhattisgarh (CH)	96.7%
5. Northeast (NE)	96.3%

Lowest Performing States (Avg Accuracy)

1. Kerala (KL)	79.2%
2. Karnataka (KT)	80.0%
3. Himachal Pradesh (HP)	83.3%
4. Andhra Pradesh (AP)	86.7%
5. Rajasthan (RJ)	86.7%

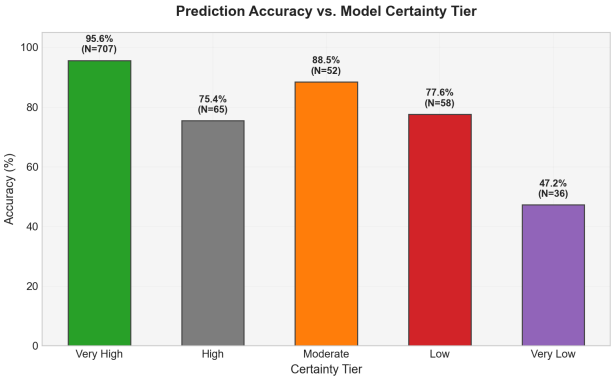
Year-Wise Accuracy Comparison Across Top 8 States



4. Certainty Analysis & Recommendations

AUDITING GUIDELINES

An analysis of prediction success against the model-assigned **Certainty Tier** reveals an excellent direct correlation. Predictions generated with **Very High Certainty** achieved a high success rate of **95.9%**. Conversely, cases categorized with *Moderate* certainty presented exactly 50% accuracy (matching random chance), indicating that these cases should be flagged for review.



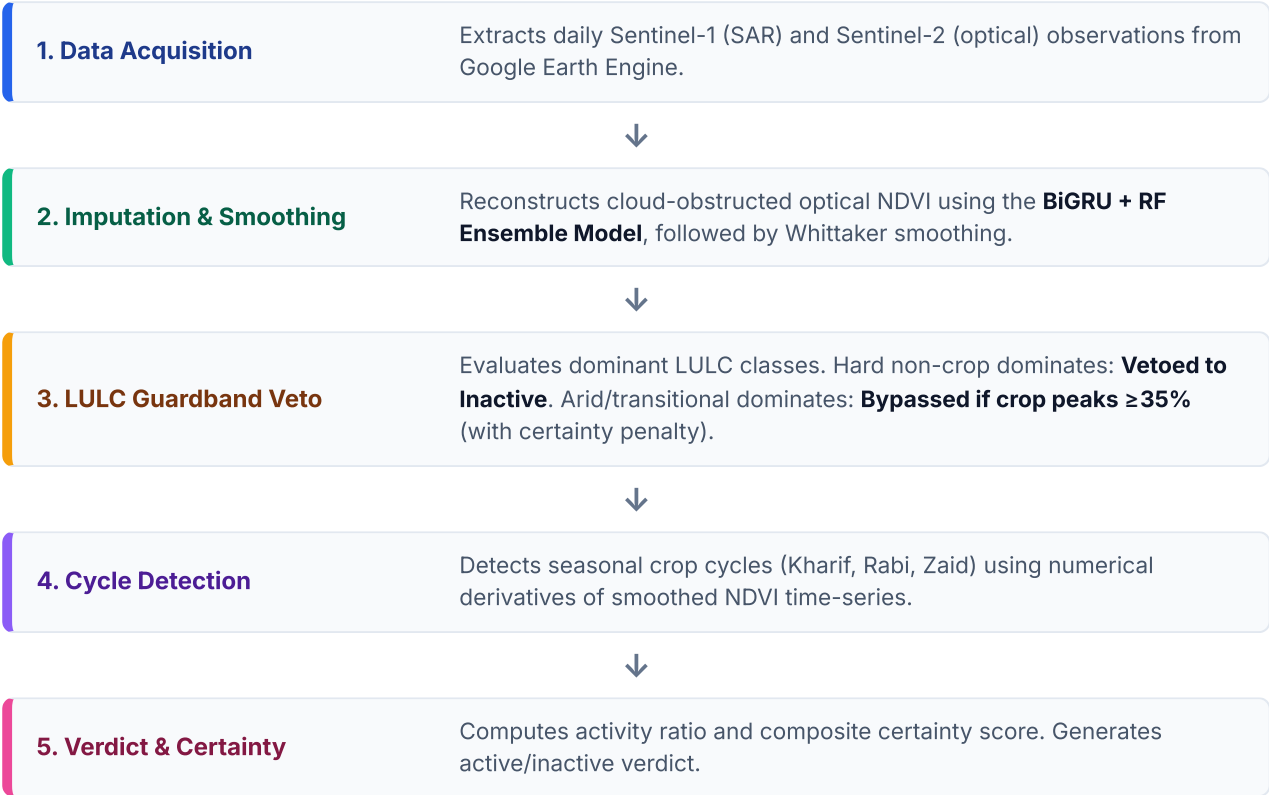
Strategic Recommendations for Audit Pipeline

- 1. Automated Certainty Gateways:** Establish a threshold where any parcel with **Moderate**, **Low**, or **Very Low** certainty is automatically routed into a manual audit pipeline. This filters out the highest-risk predictions.
- 2. Perennial Canopy Tree-Crop Filtering:** Implement tree-crop masking filters for coastal and southern regions (specifically Kerala) to prevent mature perennial plantations from distorting annual and seasonal active crop indexes.
- 3. SAR (Synthetic Aperture Radar) Integration:** Incorporate Sentinel-1 SAR imagery for states heavily impacted by monsoon cloud cover (Telangana, Karnataka, Kerala) to maintain observation frequency during active peak vegetative cycles.

5. Multi-Season Crop Activity Detection Logic

DECISION ARCHITECTURE FLOWCHART

The agricultural crop-activity detection pipeline processes co-registered multi-spectral and radar satellite observations to classify parcels and compute certainty metrics. The process is fully automated and is structured as follows:



Mathematical Certainty Scoring Logic

- Certainty represents the model's confidence in its classification. It is calculated using a composite base score scaled by three penalties:
- Missing Optical Data Penalty (0.8x):** Applied when Sentinel-2 scene count is low.
 - Transitional non-crop Dominance Penalty (0.85x):** Applied when active crop signatures are found on fields dominated by bare land, shrubs, or grass (e.g. in Rajasthan).
 - Guardband Conflict Penalty (0.5x):** Applied if LULC classifications and indices are highly contradictory.

6. Core Signal Imputation: BiGRU + RF Ensemble model

SAR-TO-OPTICAL MAPPING

Agricultural crop canopy monitoring is frequently hindered by optical cloud obstructions during key periods of the monsoon. To bridge these gaps, the pipeline uses a **BiGRU + Random Forest Ensemble Model** (60% BiGRU, 40% Random Forest) to translate cloud-penetrating Sentinel-1 Radar Vegetation Index (RVI) values to Sentinel-2 optical NDVI values.

FINAL GENERALIZATION ACCURACY (TEST R²)

0.9106

On Unseen Validation Parcels

TEST MEAN ABSOLUTE ERROR (MAE)

0.0603

Avg Deviation in NDVI units

Model Parameters & Configuration

Parameter	Setting	Description
Algorithm	BiGRU + RF Ensemble Model	Hybrid Recurrent & Decision-Tree Ensemble
Ensemble Weights	60% BiGRU / 40% Random Forest	Weighted blending of predictions
Sequence Length	5 periods (bidirectional)	Temporal context window size
BiGRU Hidden Units	64 units, 2 layers	Recurrent layer configuration
RF Estimators	100 trees	Forest base estimators
Smoothing	Whittaker filter	NDVI $\lambda=50$ RVI $\lambda=200$
EVI Status	CALCULATED	Derived from predicted NDVI (EVI = 0.665 * NDVI + 0.005)

Season-wise Generalization

Season	Test Samples	R ² Score	RMSE Error
Rabi (Winter)	2685	0.8486	0.1134
Kharif (Monsoon)	1007	0.8439	0.1223
Zaid (Summer)	1118	0.7942	0.1063

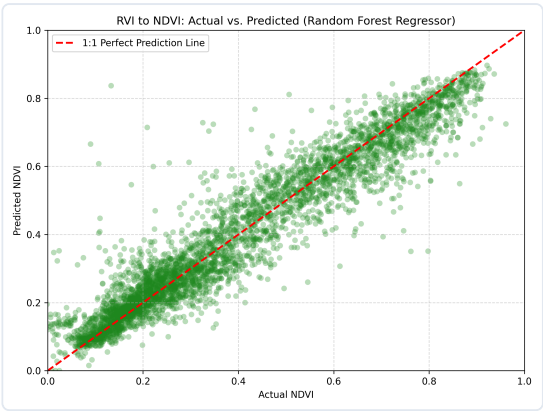


Figure 5: Predicted vs Ground-truth NDVI (Test Split)

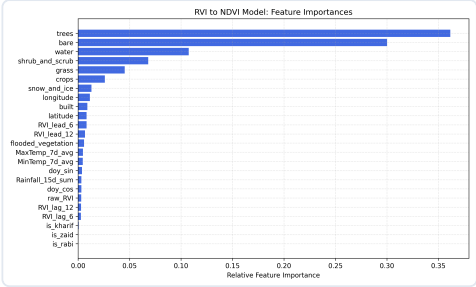


Figure 6: Model Relative Feature Importances

6. Appendix: Detailed Farm-by-Farm Reference Table

FULL CENSUS

This appendix contains a complete record of all **306 monitored farms**, detailing their ground truth classification (Static Active / Static Inactive) along with the predicted agricultural activity and certainty tier for each of the three years (2023-24, 2024-25, and 2025-26). Failed or errored GEE runs are flagged with .

Farm Name	State	Ground Truth	2023-24 (Certainty)	2024-25 (Certainty)	2025-26 (Certainty)
AP01 (Crop)	AP	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
AP02 (No-Crop)	AP	Inactive	Inactive (Very High) ✓	Inactive (Very High) ✓	Inactive (Very High) ✓
AP03 (Crop)	AP	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
AP04 (No-Crop)	AP	Inactive	Active (Very High) ✗	Active (Very High) ✗	Active (Very High) ✗
AP05 (No-Crop)	AP	Inactive	Inactive (Very High) ✓	Active (Moderate) ✗	Inactive (Very High) ✓
AP06 (Crop)	AP	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
AP07 (Crop)	AP	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
AP08 (Crop)	AP	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
AP09 (Crop)	AP	Active	Active (High) ✓	Active (Very High) ✓	Active (Very High) ✓
AP10 (No-Crop)	AP	Inactive	Active (Very High) ✗	Inactive (Very High) ✓	Inactive (Very High) ✓
AP11 (Crop)	AP	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
AP12 (Crop)	AP	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
AP13 (No-Crop)	AP	Inactive	Active (Very High) ✗	Active (Very High) ✗	Inactive (Low) ✓
AP14 (No-Crop)	AP	Inactive	Inactive (Moderate) ✓	Inactive (Low) ✓	Inactive (Very High) ✓
AP15 (Crop)	AP	Active	Active (High) ✓	Active (Moderate) ✓	Inactive (Moderate) ✗
AP16 (Crop)	AP	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
AP17 (Crop)	AP	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
AP18 (Crop)	AP	Active	Active (Very High) ✓	Active (Low) ✓	Active (Very High) ✓
AP19 (No-Crop)	AP	Inactive	Inactive (Moderate) ✓	Inactive (Very High) ✓	Inactive (Very High) ✓
AP20 (No-Crop)	AP	Inactive	Inactive (Low) ✓	Inactive (Low) ✓	Inactive (Low) ✓
BH01 (Crop)	BH	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
BH02 (No-Crop)	BH	Inactive	Active (Moderate) ✗	Inactive (High) ✓	Inactive (Very High) ✓
BH03 (No-Crop)	BH	Inactive	Inactive (Moderate) ✓	Inactive (Low) ✓	Inactive (Moderate) ✓
BH04 (Crop)	BH	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
BH05 (Crop)	BH	Active	Active (High) ✓	Inactive (High) ✗	Active (Very High) ✓
BH06 (Crop)	BH	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
BH07 (Crop)	BH	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
BH08 (Crop)	BH	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
BH10 (No-Crop)	BH	Inactive	Inactive (Low) ✓	Inactive (Low) ✓	Inactive (Moderate) ✓
CH01 (Crop)	CH	Active	Active (Very High) ✓	Active (Very High) ✓	Inactive (Very High) ✗
CH02 (No-Crop)	CH	Inactive	Inactive (Very High) ✓	Inactive (Very High) ✓	Inactive (Very High) ✓
CH03 (Crop)	CH	Active	Active (Very High) ✓	Active (Very High) ✓	Active (High) ✓
CH04 (Crop)	CH	Active	Active (Very High) ✓	Active (Moderate) ✓	Active (Moderate) ✓
CH05 (Crop)	CH	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
CH06 (Crop)	CH	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
CH07 (Crop)	CH	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
CH08 (No-Crop)	CH	Inactive	Inactive (Moderate) ✓	Inactive (Low) ✓	Inactive (Moderate) ✓
CH09 (Crop)	CH	Active	Active (Very High) ✓	Active (Very High) ✓	Active (High) ✓
CH10 (Crop)	CH	Active	Active (Very High) ✓	Active (Very High) ✓	Active (High) ✓
GJ01 (Crop)	GJ	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
GJ02 (Crop)	GJ	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
GJ03 (No-Crop)	GJ	Inactive	Active (Very High) ✗	Active (Low) ✗	Active (Very High) ✗
GJ04 (Crop)	GJ	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓

Farm Name	State	Ground Truth	2023-24 (Certainty)	2024-25 (Certainty)	2025-26 (Certainty)
GJ05 (Crop)	GJ	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
GJ06 (No-Crop)	GJ	Inactive	Active (Very High) ✗	Active (Very High) ✗	Active (Very High) ✗
GJ07 (Crop)	GJ	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
GJ08 (Crop)	GJ	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
GJ09 (No-Crop)	GJ	Inactive	Inactive (Low) ✓	Inactive (Moderate) ✓	Inactive (Low) ✓
GJ10 (Crop)	GJ	Active	Active (Very High) ✓	Active (High) ✓	Active (Very High) ✓
GJ11 (Crop)	GJ	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
GJ12 (Crop)	GJ	Active	Active (Very High) ✓	Active (High) ✓	Active (Very High) ✓
GJ13 (No-Crop)	GJ	Inactive	Inactive (Very High) ✓	Inactive (Very High) ✓	Inactive (Very High) ✓
GJ14 (Crop)	GJ	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
GJ15 (Crop)	GJ	Active	Active (Very High) ✓	Active (Low) ✓	Active (Very High) ✓
GJ16 (Crop)	GJ	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
GJ17 (No-Crop)	GJ	Inactive	Inactive (Very High) ✓	Inactive (Very High) ✓	Inactive (Very High) ✓
GJ18 (No-Crop)	GJ	Inactive	Inactive (Very High) ✓	Inactive (Very High) ✓	Inactive (Very High) ✓
GJ19 (Crop)	GJ	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
GJ20 (Crop)	GJ	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
HP01 (Crop)	HP	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
HP02 (Crop)	HP	Active	Active (High) ✓	Active (High) ✓	Active (High) ✓
HP03 (No-Crop)	HP	Inactive	Inactive (Moderate) ✓	Inactive (Moderate) ✓	Inactive (High) ✓
HP04 (No-Crop)	HP	Inactive	Inactive (Low) ✓	Inactive (Low) ✓	Inactive (Low) ✓
HP05 (Crop)	HP	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
HP06 (Crop)	HP	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
HP07 (Crop)	HP	Active	Active (Very High) ✓	Inactive (Very Low) ✗	Inactive (Low) ✗
HP08 (Crop)	HP	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
HP09 (No-Crop)	HP	Inactive	Inactive (Very High) ✓	Inactive (Very High) ✓	Inactive (Very High) ✓
HP10 (Crop)	HP	Active	Inactive (Low) ✗	Inactive (Very Low) ✗	Inactive (Very Low) ✗
HR01 (Crop)	HR	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
HR02 (Crop)	HR	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
HR03 (Crop)	HR	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
HR04 (No-Crop)	HR	Inactive	Active (Very High) ✗	Active (Very High) ✗	Inactive (Very Low) ✓
HR05 (Crop)	HR	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
HR06 (Crop)	HR	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
HR07 (Crop)	HR	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
HR08 (No-Crop)	HR	Inactive	Inactive (Very High) ✓	Inactive (Very High) ✓	Inactive (Very High) ✓
HR09 (Crop)	HR	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
HR10 (Crop)	HR	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
JH01 (Crop)	JH	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
JH02 (Crop)	JH	Active	Active (Very High) ✓	Active (Very High) ✓	Inactive (Low) ✗
JH03 (Crop)	JH	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Moderate) ✓
JH04 (Crop)	JH	Active	Active (Very High) ✓	Active (High) ✓	Active (Very High) ✓
JH05 (Crop)	JH	Active	Active (High) ✓	Inactive (Low) ✗	Inactive (Very Low) ✗
JH06 (Crop)	JH	Active	Active (Very High) ✓	Active (Very High) ✓	Active (High) ✓
JH07 (Crop)	JH	Active	Active (Very High) ✓	Active (Very High) ✓	Active (High) ✓

Farm Name	State	Ground Truth	2023-24 (Certainty)	2024-25 (Certainty)	2025-26 (Certainty)
JH08 (No-Crop)	JH	Inactive	Inactive (Very High) ✓	Inactive (Very High) ✓	Inactive (Moderate) ✓
JH09 (Crop)	JH	Active	Active (Moderate) ✓	Active (High) ✓	Active (Moderate) ✓
JH10 (No-Crop)	JH	Inactive	Inactive (Very High) ✓	Inactive (High) ✓	Inactive (Low) ✓
JK01 (No-Crop)	JK	Inactive	Inactive (Very High) ✓	Inactive (Very High) ✓	Inactive (Very High) ✓
JK02 (No-Crop)	JK	Inactive	Inactive (Very High) ✓	Inactive (Very High) ✓	Inactive (Very High) ✓
JK03 (No-Crop)	JK	Inactive	Inactive (Very Low) ✓	Inactive (Very High) ✓	Inactive (Very Low) ✓
JK04 (Crop)	JK	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
JK05 (No-Crop)	JK	Inactive	Inactive (Very High) ✓	Inactive (Very High) ✓	Inactive (Very High) ✓
JK06 (Crop)	JK	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
JK07 (Crop)	JK	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
JK08 (Crop)	JK	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
JK09 (Crop)	JK	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
JK10 (No-Crop)	JK	Inactive	Active (High) ✗	Inactive (Very Low) ✓	Inactive (Low) ✓
KL01 (Crop)	KL	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
KL02 (No-Crop)	KL	Inactive	Inactive (Very High) ✓	Inactive (Very High) ✓	Inactive (Very High) ✓
KL03 (No-Crop)	KL	Inactive	Inactive (Very High) ✓	Inactive (Very High) ✓	Inactive (Very High) ✓
KL04 (No-Crop)	KL	Inactive	Inactive (Very Low) ✓	Inactive (Low) ✓	Inactive (Very Low) ✓
KL05 (No-Crop)	KL	Inactive	Inactive (Very High) ✓	Inactive (Very High) ✓	Inactive (Very High) ✓
KL06 (Crop)	KL	Active	Inactive (Low) ✗	Inactive (Very Low) ✗	Inactive (Very High) ✗
KL07 (Crop)	KL	Active	Active (Very High) ✓	Inactive (Very High) ✗	Inactive (Very High) ✗
KL10 (No-Crop)	KL	Inactive	Inactive (Very High) ✓	Inactive (Very High) ✓	Inactive (Very High) ✓
KT01 (Crop)	KT	Active	Inactive (Low) ✗	Inactive (Very High) ✗	Inactive (Low) ✗
KT02 (No-Crop)	KT	Inactive	Inactive (Very High) ✓	Active (High) ✗	Inactive (Very High) ✓
KT03 (No-Crop)	KT	Inactive	Inactive (Very High) ✓	Inactive (Very High) ✓	Inactive (Very High) ✓
KT04 (No-Crop)	KT	Inactive	Inactive (Very High) ✓	Inactive (Very High) ✓	Inactive (Very High) ✓
KT05 (Crop)	KT	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
KT06 (Crop)	KT	Active	Inactive (High) ✗	Inactive (Very Low) ✗	Inactive (Very High) ✗
KT07 (Crop)	KT	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
KT08 (No-Crop)	KT	Inactive	Inactive (Low) ✓	Active (Low) ✗	Inactive (Low) ✓
KT09 (No-Crop)	KT	Inactive	Inactive (Very High) ✓	Active (Very High) ✗	Inactive (Very High) ✓
KT10 (Crop)	KT	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
KT11 (Crop)	KT	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
KT12 (No-Crop)	KT	Inactive	Inactive (Very High) ✓	Inactive (Very High) ✓	Inactive (Very High) ✓
KT13 (Crop)	KT	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
KT14 (Crop)	KT	Active	Active (Very High) ✓	Active (Very High) ✓	Inactive (Very High) ✗
KT15 (Crop)	KT	Active	Active (High) ✓	Active (Very High) ✓	Active (High) ✓
KT16 (Crop)	KT	Active	Active (Very High) ✓	Active (Very High) ✓	Active (High) ✓
KT17 (Crop)	KT	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
KT18 (No-Crop)	KT	Inactive	Active (Very High) ✗	Active (Very High) ✗	Inactive (Very High) ✓
KT19 (No-Crop)	KT	Inactive	Inactive (Very High) ✓	Inactive (Very High) ✓	Inactive (Very High) ✓
KT20 (No-Crop)	KT	Inactive	Inactive (Very High) ✓	Inactive (Very High) ✓	Inactive (Very High) ✓
MH01 (No-Crop)	MH	Inactive	Inactive (Very High) ✓	Inactive (Very High) ✓	Inactive (Very High) ✓
MH02 (Crop)	MH	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓

Farm Name	State	Ground Truth	2023-24 (Certainty)	2024-25 (Certainty)	2025-26 (Certainty)
MH03 (Crop)	MH	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
MH04 (Crop)	MH	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
MH05 (Crop)	MH	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
MH06 (Crop)	MH	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
MH07 (No-Crop)	MH	Inactive	Active (High) ✗	Active (High) ✗	Active (High) ✗
MH08 (Crop)	MH	Active	Inactive (Very Low) ✗	Inactive (Low) ✗	Inactive (Very Low) ✗
MH09 (Crop)	MH	Active	Inactive (Low) ✗	Active (Moderate) ✓	Inactive (Low) ✗
MH10 (Crop)	MH	Active	Active (High) ✓	Active (Very High) ✓	Active (High) ✓
MH11 (Crop)	MH	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
MH12 (Crop)	MH	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
MH13 (Crop)	MH	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
MH14 (Crop)	MH	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
MH15 (Crop)	MH	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
MH16 (Crop)	MH	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
MH17 (No-Crop)	MH	Inactive	Inactive (Very High) ✓	Inactive (Very High) ✓	Inactive (High) ✓
MH18 (No-Crop)	MH	Inactive	Inactive (Moderate) ✓	Inactive (Very Low) ✓	Inactive (Very High) ✓
MH19 (Crop)	MH	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
MH20 (No-Crop)	MH	Inactive	Inactive (Very High) ✓	Inactive (High) ✓	Inactive (Very High) ✓
MP01 (Crop)	MP	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
MP02 (No-Crop)	MP	Inactive	Inactive (Very High) ✓	Inactive (Very Low) ✓	Inactive (Very High) ✓
MP03 (Crop)	MP	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
MP04 (Crop)	MP	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
MP05 (Crop)	MP	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
MP06 (No-Crop)	MP	Inactive	Inactive (High) ✓	Inactive (Moderate) ✓	Inactive (High) ✓
MP07 (Crop)	MP	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
MP08 (No-Crop)	MP	Inactive	Inactive (Moderate) ✓	Inactive (Moderate) ✓	Inactive (Moderate) ✓
MP09 (Crop)	MP	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
MP10 (No-Crop)	MP	Inactive	Inactive (Moderate) ✓	Inactive (Moderate) ✓	Inactive (Moderate) ✓
MP11 (Crop)	MP	Active	Active (Very High) ✓	Active (High) ✓	Inactive (Very Low) ✗
MP12 (Crop)	MP	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
MP13 (Crop)	MP	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
MP14 (No-Crop)	MP	Inactive	Active (High) ✗	Active (High) ✗	Active (High) ✗
MP15 (Crop)	MP	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
MP16 (Crop)	MP	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
MP17 (No-Crop)	MP	Inactive	Inactive (Very High) ✓	Inactive (Very High) ✓	Inactive (Very High) ✓
MP18 (Crop)	MP	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
MP19 (No-Crop)	MP	Inactive	Inactive (Very High) ✓	Active (High) ✗	Active (High) ✗
MP20 (Crop)	MP	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
NE01 (Crop)	NE	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
NE02 (No-Crop)	NE	Inactive	Inactive (Very High) ✓	Inactive (Very High) ✓	Inactive (Very High) ✓
NE03 (Crop)	NE	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
NE04 (Crop)	NE	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
NE05 (Crop)	NE	Active	Active (Very High) ✓	Active (High) ✓	Active (Very High) ✓

Farm Name	State	Ground Truth	2023-24 (Certainty)	2024-25 (Certainty)	2025-26 (Certainty)
NE06 (Crop)	NE	Active	Active (Very High)	Active (Very High)	Active (Very High)
NE07 (Crop)	NE	Active	Active (Very High)	Inactive (High)	Active (Very High)
NE08 (No-Crop)	NE	Inactive	Inactive (Very High)	Inactive (Very High)	Inactive (Moderate)
NE09 (Crop)	NE	Active	Active (Very High)	Active (Very High)	Active (Very High)
OD01 (Crop)	OD	Active	Active (Very High)	Active (Very High)	Active (Very High)
OD02 (No-Crop)	OD	Inactive	Inactive (Very High)	Inactive (High)	Inactive (Very Low)
OD03 (Crop)	OD	Active	Active (Very High)	Active (Very High)	Active (Very High)
OD05 (Crop)	OD	Active	Active (Very High)	Active (Very High)	Active (Very High)
OD06 (Crop)	OD	Active	Active (Very High)	Active (Very High)	Active (Very High)
OD07 (Crop)	OD	Active	Active (Very High)	Active (Very High)	Active (Very High)
OD08 (Crop)	OD	Active	Active (Low)	Inactive (High)	Inactive (Very Low)
OD09 (No-Crop)	OD	Inactive	Inactive (Very High)	Inactive (Low)	Inactive (Very High)
OD10 (Crop)	OD	Active	Inactive (Very Low)	Active (High)	Active (High)
OD4 (Crop)	OD	Active	Active (Very High)	Active (Very High)	Active (Very High)
PB01 (Crop)	PB	Active	Active (Very High)	Active (Very High)	Active (Very High)
PB02 (Crop)	PB	Active	Active (Very High)	Active (Very High)	Active (Very High)
PB03 (Crop)	PB	Active	Active (Very High)	Active (Very High)	Active (Very High)
PB04 (No-Crop)	PB	Inactive	Inactive (Moderate)	Inactive (Low)	Inactive (Moderate)
PB05 (Crop)	PB	Active	Active (Very High)	Active (Very High)	Active (Very High)
PB06 (No-Crop)	PB	Inactive	Inactive (Low)	Inactive (Very Low)	Inactive (Low)
PB07 (Crop)	PB	Active	Active (Very High)	Active (Very High)	Active (Very High)
PB08 (Crop)	PB	Active	Active (Very High)	Active (Very High)	Active (Very High)
PB09 (Crop)	PB	Active	Active (Very High)	Active (Very High)	Active (Very High)
PB10 (Crop)	PB	Active	Inactive (Very Low)	Inactive (Very Low)	Inactive (Very Low)
PB11 (No-Crop)	PB	Inactive	Inactive (Very High)	Inactive (High)	Inactive (Very High)
PB12 (Crop)	PB	Active	Active (Very High)	Active (Very High)	Active (Very High)
PB13 (Crop)	PB	Active	Active (Very High)	Active (Very High)	Active (Very High)
PB14 (Crop)	PB	Active	Active (Very High)	Active (Very High)	Active (Very High)
PB15 (No-Crop)	PB	Inactive	Inactive (Moderate)	Inactive (Moderate)	Inactive (Moderate)
PB16 (Crop)	PB	Active	Active (Very High)	Active (Very High)	Active (Very High)
PB17 (Crop)	PB	Active	Active (Very High)	Active (Very High)	Active (Very High)
PB18 (Crop)	PB	Active	Active (Very High)	Active (Very High)	Active (Very High)
PB19 (Crop)	PB	Active	Active (Very High)	Active (Very High)	Active (Very High)
PB20 (Crop)	PB	Active	Active (Very High)	Active (Very High)	Active (Very High)
RJ01 (Crop)	RJ	Active	Active (Very High)	Active (Low)	Active (Very High)
RJ02 (Crop)	RJ	Active	Active (Low)	Active (Very High)	Active (Low)
RJ03 (Crop)	RJ	Active	Active (High)	Inactive (Moderate)	Active (High)
RJ04 (Crop)	RJ	Active	Active (Very High)	Active (Very High)	Active (Very High)
RJ05 (Crop)	RJ	Active	Active (Very High)	Active (Very High)	Active (Very High)
RJ06 (Crop)	RJ	Active	Active (Very High)	Active (Very High)	Active (Very High)
RJ07 (Crop)	RJ	Active	Active (Very High)	Active (Very High)	Active (Very High)
RJ08 (Crop)	RJ	Active	Active (Very High)	Inactive (High)	Active (Very High)
RJ09 (No-Crop)	RJ	Inactive	Inactive (Very High)	Inactive (Very High)	Inactive (Very High)

Farm Name	State	Ground Truth	2023-24 (Certainty)	2024-25 (Certainty)	2025-26 (Certainty)
RJ10 (Crop)	RJ	Active	Inactive (Very Low) ❌	Inactive (Very Low) ❌	Inactive (Very Low) ❌
RJ11 (Crop)	RJ	Active	Active (Very High) ✅	Active (Very High) ✅	Active (Very High) ✅
RJ12 (Crop)	RJ	Active	Active (Very High) ✅	Active (Very High) ✅	Active (Very High) ✅
RJ13 (No-Crop)	RJ	Inactive	Active (Very High) ❌	Active (Low) ❌	Active (Moderate) ❌
RJ14 (Crop)	RJ	Active	Active (Very High) ✅	Active (High) ✅	Active (High) ✅
RJ15 (Crop)	RJ	Active	Active (Very High) ✅	Active (Very High) ✅	Active (Very High) ✅
RJ16 (Crop)	RJ	Active	Active (Very High) ✅	Active (Very High) ✅	Active (Very High) ✅
RJ17 (Crop)	RJ	Active	Active (Very High) ✅	Active (Very High) ✅	Active (Very High) ✅
RJ18 (Crop)	RJ	Active	Active (Very High) ✅	Active (Very High) ✅	Active (Very High) ✅
RJ19 (No-Crop)	RJ	Inactive	Inactive (Very High) ✅	Inactive (Very High) ✅	Inactive (Very High) ✅
RJ20 (No-Crop)	RJ	Inactive	Inactive (Very High) ✅	Inactive (Very High) ✅	Inactive (Very High) ✅
TL01 (Crop)	TL	Active	Active (Very High) ✅	Active (Very High) ✅	Active (Very High) ✅
TL02 (Crop)	TL	Active	Active (Very High) ✅	Active (Very High) ✅	Active (Very High) ✅
TL03 (Crop)	TL	Active	Active (Very High) ✅	Active (Very High) ✅	Active (Very High) ✅
TL04 (Crop)	TL	Active	Active (Very High) ✅	Active (Very High) ✅	Active (Very High) ✅
TL05 (No-Crop)	TL	Inactive	Inactive (Low) ✅	Inactive (Very High) ✅	Inactive (Low) ✅
TL06 (Crop)	TL	Active	Active (High) ✅	Inactive (Moderate) ❌	Active (High) ✅
TL07 (Crop)	TL	Active	Active (Very High) ✅	Active (Very High) ✅	Active (Very High) ✅
TL08 (Crop)	TL	Active	Inactive (Very High) ❌	Active (High) ✅	Inactive (Very High) ❌
TL09 (Crop)	TL	Active	Active (Very High) ✅	Active (Very High) ✅	Active (Very High) ✅
TL10 (Crop)	TL	Active	Active (Very High) ✅	Active (Very High) ✅	Active (Very High) ✅
TL11 (Crop)	TL	Active	Active (Very High) ✅	Active (Very High) ✅	Active (Very High) ✅
TL12 (Crop)	TL	Active	Active (High) ✅	Active (Very High) ✅	Inactive (Very High) ❌
TL13 (Crop)	TL	Active	Active (Very High) ✅	Active (Very High) ✅	Active (Very High) ✅
TL14 (Crop)	TL	Active	Active (Very High) ✅	Active (Very High) ✅	Active (Very High) ✅
TL15 (Crop)	TL	Active	Active (Very High) ✅	Active (Very High) ✅	Active (Very High) ✅
TL16 (No-Crop)	TL	Inactive	Inactive (Very High) ✅	Inactive (Very High) ✅	Inactive (Very High) ✅
TL17 (Crop)	TL	Active	Active (Very High) ✅	Active (Very High) ✅	Active (Very High) ✅
TL18 (Crop)	TL	Active	Active (Very High) ✅	Active (Very High) ✅	Active (Very High) ✅
TL19 (No-Crop)	TL	Inactive	Inactive (Very High) ✅	Inactive (Moderate) ✅	Inactive (Very High) ✅
TL20 (Crop)	TL	Active	Active (Very High) ✅	Inactive (Very High) ❌	Active (Very High) ✅
TN01 (No-Crop)	TN	Inactive	Inactive (Very High) ✅	Inactive (Moderate) ✅	Inactive (Very High) ✅
TN02 (No-Crop)	TN	Inactive	Inactive (Low) ✅	Inactive (Low) ✅	Inactive (Low) ✅
TN03 (No-Crop)	TN	Inactive	Inactive (Very High) ✅	Inactive (Very High) ✅	Inactive (Very High) ✅
TN04 (Crop)	TN	Active	Active (Very High) ✅	Active (Very High) ✅	Active (Very High) ✅
TN05 (No-Crop)	TN	Inactive	Active (Very High) ❌	Active (Very High) ❌	Active (Very High) ❌
TN06 (Crop)	TN	Active	Active (Very High) ✅	Active (Very High) ✅	Active (Very High) ✅
TN07 (Crop)	TN	Active	Active (Very High) ✅	Active (Very High) ✅	Active (Very High) ✅
TN08 (Crop)	TN	Active	Active (Very High) ✅	Active (Very High) ✅	Active (Very High) ✅
TN09 (Crop)	TN	Active	Active (Very High) ✅	Active (Very High) ✅	Active (Very High) ✅
TN10 (Crop)	TN	Active	Active (Very High) ✅	Active (Very High) ✅	Active (Very High) ✅
TN11 (Crop)	TN	Active	Active (Very High) ✅	Active (Very High) ✅	Active (Very High) ✅
TN12 (No-Crop)	TN	Inactive	Inactive (Low) ✅	Inactive (Very High) ✅	Inactive (Moderate) ✅

Farm Name	State	Ground Truth	2023-24 (Certainty)	2024-25 (Certainty)	2025-26 (Certainty)
TN13 (No-Crop)	TN	Inactive	Inactive (Low)	Inactive (Moderate)	Inactive (Moderate)
TN14 (Crop)	TN	Active	Active (Very High)	Active (Very High)	Active (Very High)
TN15 (Crop)	TN	Active	Active (Very High)	Active (Very High)	Active (Very High)
TN16 (Crop)	TN	Active	Active (Very High)	Active (Very High)	Active (Very High)
TN17 (Crop)	TN	Active	Active (Very High)	Active (Very High)	Active (Very High)
TN18 (No-Crop)	TN	Inactive	Inactive (Very High)	Inactive (Very High)	Inactive (Very High)
TN19 (Crop)	TN	Active	Active (Very High)	Active (Very High)	Active (Very High)
TN20 (No-Crop)	TN	Inactive	Inactive (Very High)	Inactive (Moderate)	Inactive (Very High)
UK01 (Crop)	UK	Active	Active (Very High)	Active (Very High)	Active (Very High)
UK02 (Crop)	UK	Active	Active (Very High)	Active (Very High)	Inactive (Very Low)
UK03 (No-Crop)	UK	Inactive	Inactive (Low)	Inactive (Moderate)	Inactive (Low)
UK04 (No-Crop)	UK	Inactive	Inactive (High)	Inactive (Very High)	Inactive (High)
UK05 (Crop)	UK	Active	Active (Very High)	Active (Very High)	Active (Very High)
UK06 (Crop)	UK	Active	Active (Very High)	Active (Very High)	Active (Very High)
UK07 (No-Crop)	UK	Inactive	Inactive (Moderate)	Inactive (Low)	Inactive (Low)
UK08 (Crop)	UK	Active	Active (Very High)	Active (Very High)	Active (Very High)
UK09 (Crop)	UK	Active	Active (Very High)	Active (Very High)	Active (Very High)
UK10 (Crop)	UK	Active	Active (Very High)	Active (Very High)	Active (Very High)
UP1 (Crop)	UP	Active	Active (Very High)	Active (Very High)	Active (Very High)
UP10 (Crop)	UP	Active	Active (High)	Active (Low)	Active (High)
UP11 (Crop)	UP	Active	Active (Very High)	Active (Very High)	Active (Very High)
UP12 (Crop)	UP	Active	Active (Very High)	Active (Very High)	Active (Very High)
UP13 (Crop)	UP	Active	Active (Very High)	Active (Very High)	Active (Very High)
UP14 (Crop)	UP	Active	Active (Very High)	Active (Very High)	Active (Very High)
UP15 (No-Crop)	UP	Inactive	Inactive (Very High)	Inactive (Very High)	Inactive (Very High)
UP16 (Crop)	UP	Active	Active (Very High)	Active (Very High)	Active (Very High)
UP17 (Crop)	UP	Active	Active (Very High)	Active (Very High)	Active (Very High)
UP18 (Crop)	UP	Active	Active (Very High)	Active (Very High)	Active (Very High)
UP19 (Crop)	UP	Active	Active (Very High)	Active (Very High)	Active (Very High)
UP2 (Crop)	UP	Active	Active (Very High)	Active (Very High)	Active (Very High)
UP20 (Crop)	UP	Active	Active (Very High)	Active (Very High)	Active (Very High)
UP3 (Crop)	UP	Active	Active (Very High)	Active (Very High)	Active (Very High)
UP4 (Crop)	UP	Active	Active (Very High)	Active (Very High)	Active (Very High)
UP5 (No-Crop)	UP	Inactive	Inactive (Very Low)	Inactive (Very Low)	Inactive (Very Low)
UP6 (Crop)	UP	Active	Active (Moderate)	Active (Low)	Active (Moderate)
UP7 (No-Crop)	UP	Inactive	Inactive (Low)	Inactive (Very Low)	Inactive (Low)
UP8 (Crop)	UP	Active	Active (Very High)	Active (Very High)	Active (Very High)
UP9 (Crop)	UP	Active	Active (Very High)	Active (Very High)	Active (Very High)
WB01 (No-Crop)	WB	Inactive	Inactive (Very Low)	Inactive (Moderate)	Inactive (Very Low)
WB02 (Crop)	WB	Active	Active (Very High)	Active (Very High)	Active (Very High)
WB03 (Crop)	WB	Active	Active (Very High)	Active (Very High)	Active (Very High)
WB04 (Crop)	WB	Active	Active (High)	Active (Moderate)	Inactive (High)
WB05 (Crop)	WB	Active	Active (Very High)	Active (Very High)	Active (Very High)

Farm Name	State	Ground Truth	2023-24 (Certainty)	2024-25 (Certainty)	2025-26 (Certainty)
WB06 (No-Crop)	WB	Inactive	Inactive (Moderate) ✓	Inactive (Moderate) ✓	Inactive (Very High) ✓
WB07 (Crop)	WB	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very Low) ✓
WB08 (Crop)	WB	Active	Active (Very High) ✓	Active (Very High) ✓	Active (Very High) ✓
WB09 (Crop)	WB	Active	Active (Low) ✓	Active (High) ✓	Inactive (Very Low) ✗
WB10 (Crop)	WB	Active	Active (Very High) ✓	Active (Very High) ✓	Active (High) ✓